



Predicting Song Popularity Using Spotify Audio Features

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Introduction

Quantifiable music characteristics such as the energy, acousticness, danceability, and valence of songs are used to manage music streaming platforms such as Spotify. This project aimed to explore the possibility that quantifiable music characteristics could explain the popularity of songs. Song popularity was considered the response variable, whereas the predictor variables were the audio characteristics such as danceability, energy, loudness, tempo, and valence. It is important to know which audio features drive song popularity because it has real-world implications for artists, producers, and as well as streaming platforms. Additionally, it can have implications for music productions decisions and improve recommendation algorithms.

Methods

Data is obtained from the Spotify Web API (18,835 songs), a reliable and standardized source of audio features
The Data is used for academic purposes in compliance with Spotify policies
Observational unit: individual song

Response variable: song popularity
○ (score from 0 to 100)

The key predictors are summarized in table below
Some additional predictors considered: acousticness, speechiness, instrumentalness, liveness, duration

Outliers were detected using the IQR method and removed if extreme
Categorical variables (key, mode, time signature) converted into dummy variables

Data was split into training (75%) and testing (25%) sets
Multiple linear regression used to model song popularity
Forward stepwise selection used to identify important predictors
Model performance was evaluated using RMSE on the test set

Variable	Meaning	Unit	Range	Median
Song Popularity	The measure of a song's popularity	Score	0-100	56
Danceability	How suitable the song is for dancing?	Index	0-0.99	0.65
Energy	The perceptual measure of song intensity/activity	Index	0-1.00	0.67
Tempo	The speed of the song	BPM	0-242	120
Loudness	The overall loudness of the song	Decibels	-38.77 - 1.59	-6.56

Exploratory Results

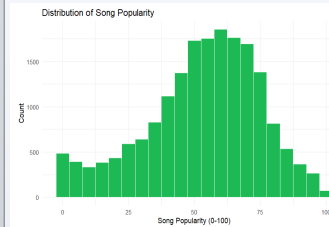


Figure 1

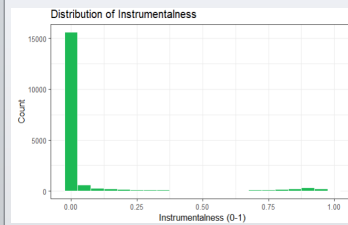


Figure 2

1: Song popularity distribution is approximately bell-shaped and centered on 56, with a slight left skew and bimodal with small concentration on 0. Most songs are classified as moderately popular and very few exist as extreme examples of either high or low popularity indicating there isn't much chance to find songs with polar opposite levels of popularity.

2: The distribution of the instrumentalness attribute is extremely right skewed, showing that the instrumentalness value of most songs is near zero. It means that there are many songs that contain vocals, and fewer songs that are instrumentals. There are some songs whose instrumentalness values are quite high.

3: The two features exhibit a slight negative linear relationship, meaning that the higher the instrumentalness, the lower the song popularity. It becomes evident when looking at the scatter plot because highly popular songs have relatively low instrumentalness.

4: The popularity distributions of songs in major or minor keys are almost identical, with the median being approximately 57 in both cases; thus, there is little effect of the mode on the popularity of a song. Major (n = 11831), Minor (n = 7004).

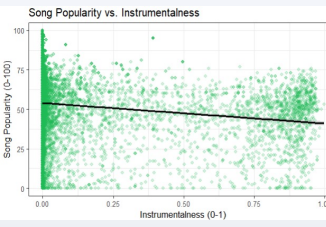


Figure 3

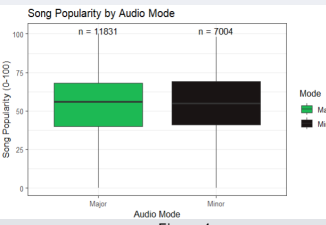


Figure 4

Predictive Results

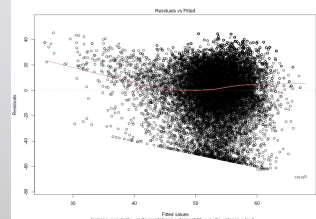


Figure 5

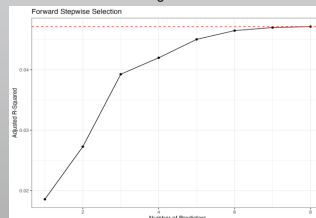


Figure 6

5. Multiple linear regression was used to predict song popularity based on selected Spotify audio features. This model assumes linearity, independence, normality, and constant variance. The residuals vs. fitted plot shows slight curvature and a mild funnel shape, suggesting some violation of linearity and evidence of heteroscedasticity.

6. A forward stepwise selection approach was used to choose predictors based on improvements in adjusted R-squared. The final model includes instrumentalness, danceability, audio_valence, loudness, energy, acousticness, liveness, and tempo. While adjusted R-squared increases with more predictors, the gains become smaller, indicating diminishing returns.

7. Cross-validation was used to evaluate model performance across different model sizes using 5-fold splits. The metric used was mean squared error. The plot shows that error decreases as predictors are added, but improvements become minimal, suggesting that additional variables provide limited benefit.

8. A Q-Q plot was used to assess the normality of residuals. Ideally, points should follow the reference line. Here, deviations at the tails indicate that residuals are not perfectly normal, suggesting a mild violation of the normality assumption.

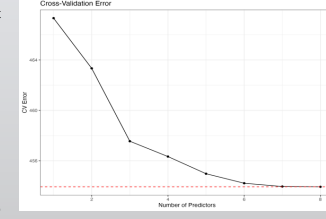


Figure 7

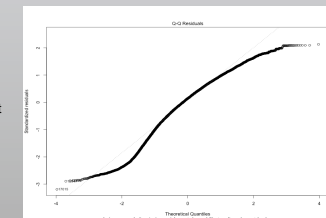


Figure 8

Discussion

As the adjusted R-squared of the fitted model is higher than the null model's R-squared, and the testing RMSE of the fitted model is also smaller (21.73) than the testing RMSE of the null model (22.17), it can be stated that the fitted model performed better than the null model. However, the difference between the two models is relatively small, indicating limited practical significance despite the improvement.

Instrumentalness, danceability, audio valence, loudness, energy, acousticness, liveness, and tempo were included as predictors in the final model. While these variables contributed to predicting song popularity, the model had low explanatory power with an adjusted R-squared of approximately 0.047. Additionally, cross-validation using 5 folds showed only marginal reductions in prediction error as predictors were added, further supporting that the model provides limited predictive improvement.

There were also technical challenges related to violations of linear regression assumptions, including non-constant variance and non-normal residuals, as observed in the residuals vs. fitted plot and Q-Q plot. Although the overall ANOVA results were statistically significant ($F \approx 88.33$, $p < 0.001$), indicating that at least one predictor is useful, these assumption violations and the low R-squared suggest caution when interpreting the model.

It is important to consider the broader implications of this model, as it may reinforce existing popularity trends in recommendation systems. This means that already popular songs could be promoted more frequently, potentially limiting exposure for less popular content.

Metric	Null Model	Fitted Model
Adjusted R-Squared	0.000	0.047
Residual Standard Error	21.82	21.3
ANOVA P-value	NA	< 0.001
Cross-Validated Error	22.17	21.73

Conclusions

In this analysis, several independent variables were considered in a multiple linear regression model to predict song popularity based on audio features. The results show some predictive ability, though overall accuracy remains limited. Beyond adding more predictors, the model could be improved by addressing violations of assumptions (e.g., transforming variables), incorporating interaction or nonlinear terms, or exploring more flexible models such as regularization or tree-based methods.